

AI USED IN SELF DRIVING CAR

How Ai is used ?

Artificial intelligence (AI) is playing an ever-increasing role in the automotive industry and is considering the key to autonomy for self-driving cars. The human being is remarkable in that it can drive vehicles at high speed and in urban areas with many things happening on the road and to the side – and yet a person can, usually, manage all these visual and aural cues without causing an accident.

These things are very challenging for a computer. For all these inputs, such as traffic lights, pedestrians, other cars on the road, roadworks, and challenges such as rain, fog and snow, the vehicle needs to replicate the ability of the human brain to sense what is happening. It needs to anticipate and predict what will happen next, and then recommend a course of action. The AI has to decide on whether to slow the car down by applying the brake or to drive carefully to avoid an obstacle.

Benefits

Improved road safety and reduced accidents: Human errors, such as fatigue, sleep, and distraction, are the leading causes of most accidents.

AI-powered self-driving vehicles can significantly reduce accidents and incidents resulting from human errors. With adaptive capabilities, visual capabilities, and proximity sensing analysis, AI allows AVs to operate safely and avert dangerous scenarios.

Increased mobility: AVs can change how the disabled, elderly, and people with medical conditions travel. Using speech recognition or mobile apps, the vehicle can safely navigate these groups to their destination. Moreover, ridesharing robotaxis can provide an affordable and convenient alternative to traditional modes of transport.

Reduced traffic congestion and environmental impact: AVs can detect and avoid traffic using dynamic routing capabilities, route planning, and maneuvering. AI in AVs can manage energy by optimizing braking and acceleration, limiting fuel consumption. This eco-friendly strategy reduces greenhouse emissions.

Improved traffic management: Self-driving cars using V2X frameworks can communicate with other road agents to understand traffic conditions. This information can help in route planning and avoiding congested areas.

Challenges

Technical limitations and edge cases: Some of the technical complexities facing the advancement of AVs include limitations in onboard computing power, perception accuracy, localization challenges, and hardware. Simulating rare edge cases that can occur in the real world is also challenging.

Regulatory hurdles and liability concerns: There is a lack of clarity on who is liable if an AV gets into an accident. Additionally, it is challenging to adapt existing laws to work for self-driving vehicles. To mitigate this, stakeholders must develop a comprehensive framework to manage liability. By following

standardized frameworks, lawmakers can establish laws prioritizing human safety and encouraging innovation.

Consumer trust and acceptance: Self-driving cars come under sharp scrutiny every time they are involved in accidents. For instance, Cruise robotaxis faced criticism in San Francisco following two separate accidents. Such incidents only continue to build mistrust in the community with safety concerns.

Companies must prioritize user safety in design to boost consumer acceptance and perform vigorous testing.

Ethical dilemmas: AI systems in AVs may face morally challenging situations, forcing them to choose between ambiguous situations. For instance, choosing between protecting onboard passengers or pedestrians in case of an accident. Stakeholders must develop acceptable frameworks and standards to tackle some of the more pressing moral dilemmas.

Future Scope

The last decade has witnessed a surge in the development of self-driving technology, propelled by breakthroughs in artificial intelligence (AI), machine learning (ML), and deep learning (DL). These cutting-edge technologies are empowering engineers to address complex challenges surrounding scene perception, motion control, path planning, and behaviour arbitration in autonomous vehicles.

The Society of Automotive Engineers (SAE) has established a framework that defines six levels of driving automation, ranging from level 0 to level 5. As the levels progress, the need for human intervention diminishes.

A Level 1 vehicle has a single feature, such as cruise control, to manage its speed.

At level 2, assistive technologies like adaptive cruise control, lane departure warning, and lane-keeping systems employ AI.

Level 3 relies heavily on AI for navigation and decision systems, allowing vehicle systems more control. It also features a suite of sensors and devices for data collection. Only a handful of companies in Europe and North America, publicly offer this level of autonomy with authorisation to drive.

Level 4 systems can do almost everything Level 5 systems can do, though they're limited by geographic and weather conditions, where human drivers need to take over.

Level 5 representing fully autonomous vehicles that require no human input or oversight

The future of AI in autonomous vehicles is shifting from basic environment perception to **superhuman intelligence**, focusing on edge-based **neural networks**, **Explainable AI (XAI)** to ensure safety transparency, and **Vehicle-to-Everything (V2X)** to prevent accidents.

The future of AI in this sector encompasses several key areas:

- **Vehicle-to-Everything (V2X) Communication:** AI will empower vehicles to talk to smart traffic lights, pedestrians' smartphones, and other connected cars to form a cooperative traffic network, virtually eliminating blind-spot accidents
- **Predictive Behaviour Modelling:** Moving beyond simple object detection, AI will use predictive analytics to anticipate the micro-movements of pedestrians and human drivers, reacting to hazards milliseconds before they fully materialize.

- **Explainable AI (XAI):** To build consumer and regulatory trust, AI systems will evolve to "explain" their driving decisions in real-time, mapping out why the car chose to brake, swerve, or merge.
- **Edge Computing and Local Processing:** Instead of relying on cloud data centres, AI models will run on localized, dedicated neural network accelerators directly in the vehicle, ensuring immediate reaction speeds even in areas without internet connectivity.
- **AI-Enhanced Fleet Logistics:** The commercial sector will see massive growth with AI managing autonomous 24/7 trucking and last-mile delivery robots, significantly reducing supply chain costs.

What latest technology used in this field ?

Self-driving cars rely on a combination of Computer Vision, **sensor fusion**, and **deep learning** to replicate human perception and decision-making. AI models process real-time data to identify objects, predict behaviours, and execute safe driving commands.

The core technologies driving this AI system include:

- **Computer Vision:** Gives the vehicle "eyes" by analysing camera feeds in real-time to detect traffic lights, signs, lane markers, and pedestrians.
- **Deep Learning & Neural Networks:** Uses layers of algorithms (like Convolutional Neural Networks) to analyse complex video data, recognize patterns, and make split-second driving decisions.
- **Sensor Fusion:** Combines data from various hardware inputs (cameras, LiDAR, radar, and ultrasonic sensors) to create a single, accurate 3D map of the car's surroundings.
- **Behaviour Prediction:** AI models analyse the trajectory and speed of surrounding pedestrians and vehicles to anticipate their next moves, such as merging or stopping.

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